

Assignment 2

The output when running unprotected can be seen below

```
Val P1: 30
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 14.0
Val P2: 17
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 8.0

Val P1: 32
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 15.0
Val P2: 18
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 9.0

Val P1: 33
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 16.0
Val P2: 19
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 9.5

Val P1: 35
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 17.5
Val P2: 20
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 10.0

Val P1: 37
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 18.5
Val P2: 21
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 10.0

Val P1: 39
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 19.0
Val P2: 22
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 11.0

Val P1: 40
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 20.0
Val P2: 23
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 11.0

Val P1: 42
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 20.5
Val P2: 24
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 12.0

Val P1: 44
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 21.5
Val P2: 25
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 12.0

Val P1: 46
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 22.0
Val P2: 26
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 13.0

Val P1: 47
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 23.0
Val P2: 27
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 13.5

Val P1: 49
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 24.0
Val P2: 28
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 14.0
```

Here the values are sometimes changed after the value is checked to be even, leading to halve values in the final string. To make this noticeable a delay was used in between checking and writing out the numbers.

As seen in the screenshot below the halve values disappear in the protected version.

Matthias Ellinger

```
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Val P2: 21
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 10.0

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Val P1: 41
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Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 14.0

Val P1: 49
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 24.0
Val P2: 29
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 14.0

Val P1: 51
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 25.0
Val P2: 30
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 15.0

Val P1: 52
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 26.0
Val P2: 31
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 15.0

Val P1: 54
Str P1: This is a test message to show the importance of task synchronisation. The current value of sensor 1 reads 26.0
Val P2: 32
Str P2: This message belongs to aircraft Airbus A350 flight number GFIK42. The current value of sensor 2 reads 16.0
```

Defining PROTECTED to use the protected version.